

# CONCENTRATION, NOT UNCERTAINTY: WHY TARGETED SYNTHETIC DATA DOESN'T HELP CAMOUFLAGED OBJECT DETECTION

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## ABSTRACT

Camouflaged object detection asks a model to outline objects that blend into their background. Labels are expensive, tracing a boundary you can barely see is slow, so synthetic camouflaged images are a common partial remedy. However, which images should you generate?

We built and tested an answer: cluster the unlabelled real images the model will face, find the clusters it is least sure about, generate images aimed there, retrain, repeat. The ingredients are standard: active learning, mean-teacher disagreement, and clustering, but the loop is ours.

No advantage survives measurement. Across 36 controlled training runs, aiming the budget by uncertainty does not beat spending it at random at any effect size this design resolves: the gap sits inside the seed-to-seed spread on both architectures, and an arm-specific interval places it in  $[-0.004, +0.014]$ , excluding the improvement reported for this approach but not effects several times smaller. Two kinds of evidence sit underneath that sentence and should not be conflated. The *geometric* evidence is strong and positive: concentration, not the uncertainty direction, is what makes a targeted training set measurably different from a random one, and we show this by a control that keeps our allocation's shape while destroying the signal inside it. The *downstream* evidence is a null under stated sensitivity: the signal's direction moves trained accuracy by nothing we can resolve. On boundary metrics, which is what the acquisition signal was built to move, the concentration effect does become resolvable, and the anti-targeted arm captures it just as fully.

Independently: 50 of CHAMELEON's 76 images (65.8%) are re-encoded, rescaled, or cropped copies of COD10K and CAMO training images — 41 reproducible by a released detector, 9 more confirmed pixel-by-pixel after geometric alignment. Sixteen further images resolve neither way, so 50 is a floor rather than an estimate. Exact hashing, the standard check, reports zero overlap because the copies were re-saved in a different format. We claim non-independence rather than inflation, and release a detector and a clean protocol. A control that returns zero is informative only if it could have returned a different value.

## 1 INTRODUCTION

Camouflaged object detection asks for a pixel-accurate outline of an object that is hard to see on purpose. Its training masks are unusually expensive, because an annotator must decide where a boundary lies when the point of the object is that its boundary is invisible. One response to the resulting scarcity is to manufacture more: a generator can place a real foreground object into a freshly generated background. But generation is not free and every generated image has to be trained on, so if you can make a thousand images, which thousand?

**Our answer.** Let the model choose: train it, find the unlabelled real images it is least sure about, aim the generation there, retrain, repeat. Every ingredient is standard — uncertainty sampling from active learning Settles (2009), a mean-teacher disagreement score Tarvainen & Valpola (2017), clustering over self-supervised features Oquab et al. (2024); Radford et al. (2021) so the budget spreads rather than piling onto near-duplicates. The composition is ours, and we could not make it work.

**What we found, and how far the measurement reaches.** Across 24 training runs — two architectures, four conditions, three seeds — a targeted allocation does not beat a random one: the decisive gap is WITHIN NOISE on both architectures and both endpoints. The pre-registered bar is *pooled* across four arms and so dominated by the unpadded arm A0, which is not in the comparison at all; at 126% of the improvement the published approach reports, it could not have detected that improvement. A second campaign of 12 runs tightens it by  $3.24\times$ , leaving the decisive gap at  $0.91\times$  threshold. An interval on the two arms actually compared is tighter than either:  $\Delta(C10 - B) = +0.0050$ , 95%  $[-0.0035, +0.0136]$ , which *excludes* the 0.0142 reference improvement while admitting effects up to about two-thirds of it (Section 4). All three readings say *no effect resolvable at this sensitivity*, never a demonstration of absence. A mean-teacher positive control shows the instrument is not blind, detecting a real effect of  $+0.0197$  at  $3.3\times$  its bar — though not, as Section 4 states, an effect the size of the nulls.

**Why it looks like it works, and a second finding.** Targeted selection really does produce a training set measurably different from a random one — but so does any rule that concentrates the budget on a few clusters, including rules that keep our allocation’s exact shape while destroying the uncertainty signal inside it. What looked like the effect of aiming was the effect of concentrating, so the discriminating control is not random selection but a same-shape shuffle. And the concentration effect is not only geometric: on boundary metrics — the quantity our acquisition signal was designed to move — concentration produces a difference that clears its bar on trained accuracy, while the *anti*-targeted arm captures that difference just as fully (Section 4.2). Separately: the CHAMELEON benchmark is more than half training data, and the standard check returns clean. Some of what follows is specific to the loop we built and some travels; we say which for each. One pattern runs through both — a control that returns zero is informative only if it could have returned something else.

**Contributions.** The first is independent of our method, the next three travel beyond our setup, and the last is a property of the loop we built.

- 50 of CHAMELEON’s 76 images (65.8%) are re-encoded, rescaled or cropped COD10K-train and CAMO training images, while exact hashing returns 0. A further 16 resolve neither way, so 50 is a floor and no image was counted on suspicion. A released detector reproduces 41 of them; the remaining 9 leave their dimension group and are confirmed by warping each candidate into its partner’s frame and measuring the residual, which separates a rescale or crop from a second photograph of the same specimen. Verified against an author-sourced release, calibrated to 41/41 recall on the confirmed pairs, and held to a negative control that flags 0 of 120 on NC4K. We claim non-independence rather than inflation, and identify a published protocol whose unlabelled target pool holds 44 of the 76 test images (Section 5.2).
- The target distribution separates only weakly into clusters under every embedding we tested — best silhouette 0.1600 across three spaces — and the budget a cluster-wise policy then spends is aimed largely at one component of a two-dataset mixture: 51.5% of it lands on CAMO images, which are 24.8% of the pool, and the two largest quotas take a third of the budget while holding 7 of the 2026 primary-endpoint images (Section 4.3).
- An uncertainty signal can contribute nothing resolvable beyond the concentration it induces:  $+0.0073$  of a Cohen’s  $d$  against its own shuffle, and with concentration held *exactly* fixed, real, destroyed and reversed targeting are indistinguishable in 12/12 cells. Where the concentration effect *is* resolvable on trained accuracy — boundary metrics on SINet — the reversed arm reproduces it in full (Sections 4.1, 4.2).
- Uncertainty predicts pixel error better than structural error in six independent signal-architecture tests, surviving an area control that clears its floor by no margin on one of two covariates (Section 4.4).
- What would have to change before this loop could show a gain: a pinned optimisation budget under which added images displace rather than augment, a foreground supply that is a bijection onto 4447 copied objects, and a generated distribution narrower than its own input; we also *withdraw* one of our own candidate causes (Section 5.1). None licenses the conclusion that closed-loop generation cannot work.

## 2 RELATED WORK

**Scope and closest systems.** Camouflaged object detection (COD) requires pixel-level segmentation of objects whose colour and texture closely match their surroundings; Fan et al. (2023) provide a broad account of its architectures, datasets, and evaluation conventions. This paper addresses a different question: whether an *acquisition policy* can improve a fixed synthetic-data pipeline. Synthetic augmentation methods such as DatasetDM Wu et al. (2023) address how to construct labelled training data, whereas our loop asks which generated examples to add under a fixed budget. The closest systems are LAKE-RED Zhao et al. (2024), whose background retrieval and compositing we use unchanged, and S2R-COD Luo et al. (2025), whose LAKE-RED-based synthetic-to-real self-training protocol we follow. Relative to these systems, the intervention under test is the uncertainty-guided allocation layer: it ranks target clusters using student–teacher disagreement and sends the resulting quotas to the generator. The comparison of C10 with B therefore tests targeting against random allocation while keeping the generator and budget fixed; the subsequent same-shape shuffle tests whether any difference is attributable to the uncertainty direction rather than concentration (Section 4.1).

**Acquisition and uncertainty.** The design draws on active learning Settles (2009), but its query is directed to a synthetic generator rather than a human annotator. Mean-teacher consistency Tarvainen & Valpola (2017) supplies the unlabelled disagreement signal, and  $k$ -means over fixed self-supervised features implements a lightweight diversity constraint in the spirit of core-set selection Sener & Savarese (2018). Deep ensembles Lakshminarayanan et al. (2017), MC-Dropout Gal & Ghahramani (2016), and BADGE Ash et al. (2020) represent alternative uncertainty or uncertainty–diversity constructions, but are not evaluated here. The contribution is consequently the controlled test of this composition, not a new detector, generator, or uncertainty estimator.

**Uncertainty-guided synthesis, curricula, and dataset valuation.** Three lines of work share our premise that a generation budget should be steered rather than spent uniformly; Table 1 places them beside us. GAUDA Frisch et al. (2025) is closest in spirit — it conditions a latent diffusion model on the classes a Bayesian model is most uncertain about, synthesising online, and reports gains on surgical segmentation — but its generator can synthesise the object as well as its context, its signal is class-wise rather than cluster-wise, and its budget is not fixed in advance. DisCL Liang et al. (2025) steers image-guidance strength, controlling *proximity* explicitly where our loop controls only which foregrounds are drawn. SynQuE Chen & Zhong (2025) formalises the complementary problem of ranking synthetic datasets from unannotated real data; its proxy scores are the kind of quantity Section 4.1 finds is driven by concentration. **None of the three runs a signal-destroying control at fixed allocation shape**, which is the control we argue is discriminating.

**Seed-level inference, equivalence, and contamination auditing.** A null over training runs is a statement about a distribution over seeds, so we report seed-paired differences against a pre-registered spread rather than significance Pineau et al. (2021), supplemented post-hoc by equivalence intervals in the sense of Lakens (2017) — which at  $n = 3$  we report as uninformative rather than as support. On auditing, near-duplicate analyses of CIFAR and ImageNet establish that visually identical images cross split boundaries Barz & Denzler (2020), and contamination-aware construction is standard in language-model evaluation Wu et al. (2024); what is missing for image benchmarks is a check tolerant of *re-encoding* and rescaling. Appendix A covers benchmark definitions, architectural baselines and generative alternatives.

**Evaluation context.** We retain the standard COD setting summarized by Fan et al. (2023): COD10K and NC4K provide the principal evaluation context, while CAMO and CHAMELEON are established complementary benchmarks. SInet and SInet-v2 are used as fixed off-the-shelf detectors, and the reported S-measure, E-measure, and weighted  $F_{\beta}^w$  follow the conventions of the published synthetic-to-real baseline. These choices make the experiment a test of allocation under the existing COD protocol, rather than a comparison among detector architectures.

Table 1: Where this work sits among systems that steer a synthetic-data budget. *Acquisition signal* is what decides where the budget goes; *generator control* is what the method can actually vary in the generated image; *diversity control* is what stops the budget piling onto near-duplicates; *closed-loop* means the downstream model’s state feeds back into generation; *abstention / control* is whether a signal-destroying or same-shape control is run, which is what separates an effect of aiming from an effect of concentrating. The last column is the one this paper argues is missing from the literature, not a claim of superiority — our own entry records a null.

| Work                        | Task            | Acquisition signal        | Generator control | Closed-loop   | Abstention / control                             |
|-----------------------------|-----------------|---------------------------|-------------------|---------------|--------------------------------------------------|
| LAKE-RED Zhao et al. (2024) | COD synthesis   | none (uniform)            | background only   | no            | —                                                |
| S2R-COD Luo et al. (2025)   | COD adaptation  | none (uniform)            | background only   | pseudo-labels | —                                                |
| GAUDA Frisch et al. (2025)  | surgical seg.   | epistemic, per class      | image + mask      | yes, on-line  | none reported                                    |
| DisCL Liang et al. (2025)   | long-tail cls.  | hardness, per sample      | guidance strength | curriculum    | none reported                                    |
| SynQuE Chen & Zhong (2025)  | dataset ranking | proxy score, per set      | n/a (picks sets)  | no            | none reported                                    |
| <b>This work</b>            | COD adaptation  | disagreement, per cluster | background only   | pseudo-labels | <b>same-shape shuffle, rank reversal, oracle</b> |

### 3 METHODS

The idea is easy to picture. Train a model on the labelled images you have; look at the unlabelled real images it will be judged on and find the ones it handles least confidently; ask a generator for new images resembling those, add them, and train again. What follows is our instantiation, specified exactly, because a null is only as informative as the thing it tested.

#### 3.1 EXPERIMENTAL SETTING AND CLOSED-LOOP PROTOCOL

The setting is source-to-real domain adaptation: we train on a *source pool* of 4447 labelled image and mask pairs and want the model to work on a *target pool* of 4040 unlabelled real images, with verdicts read from two held-out *endpoints*, COD10K-test primary and NC4K secondary. The target pool is not one distribution — it mixes 3040 COD10K images with 1000 from CAMO, separable by a held-out probe at up to 0.9225 AUC — so a method partitioning it is partitioning a mixture (Section 4.3).

Training runs in two rounds. In round one a student trains on the source pool while a teacher — a slowly updated running average of the student’s weights — gives a second opinion on the unlabelled images that the student must agree with. Between rounds, target images whose student–teacher edge disagreement falls below a fixed fraction of the set average, meaning the two already agree, are appended with the teacher’s prediction as their label. Round two rebuilds both models from ImageNet initialisation and trains on the enlarged pool; its best teacher checkpoint is evaluated. Two properties matter and neither is visible from “closed loop”: the round count is a fixed constant, not a convergence criterion, and the generator is never invoked from training but renders its pool once, offline, the only returning edge carrying pseudo-labels rather than images. It composites rather than synthesises (Section 5.1).

#### 3.2 UNCERTAINTY-GUIDED DATA ALLOCATION

Our allocation spends a fixed generation budget where the model is weakest, in four stages.

(i) **Group the target images**, so “weak” can name a region of the data rather than one image:  $k$ -means into  $k = 75$  clusters over L2-normalised self-supervised embeddings, fitted once and never

refitted. **(ii) Score each image for uncertainty.** We need a number, computable without labels, saying how unsure the model is: how far the student and its running average disagree. The consistency loss already reduces that quantity, which is what makes it available on unlabelled data:

$$ES(x) = a \|\nabla\sigma(s(x)) - \nabla\sigma(t(x))\|_1 + b \text{BCE}(\sigma(s(x)), \sigma(t(x))), \quad (1)$$

with student and teacher logits  $s, t$ , sigmoid  $\sigma$ , Sobel gradient magnitude  $\nabla$ , and configured weights  $a = 0.9, b = 0.3$ ; a cluster’s score  $es_j$  is the mean over its members. **(iii) Convert scores into a budget split.** Budget  $B$  is spread by a temperature softmax over  $es_j$  with  $T = \alpha \cdot \text{sd}(es)$ , at  $\alpha = 1.0$  and  $B = 1000$ ; the temperature sets how sharply the budget piles onto the worst clusters. **(iv) Fill each quota.** A funded cluster draws images by cosine similarity between its centroid and a grey-masked cutout of each candidate foreground. Appendix B gives the rest.

### 3.3 EXPERIMENTAL CONDITIONS

Four *arms* — training conditions — share one base pool and differ only in what is appended. **A0**, the unpadded baseline, is the base pool of 4447 pairs alone — an *unclean* control, differing in pool size as well as content, and *not* the configuration corresponding to the published row, which is C10 (Section 4). **A2** adds the 1000 original photographs of arm B’s stems, separating pool size from generated content. **B** adds 1000 renders drawn uniformly at random, redrawn per seed; **C10** adds the 1000 chosen by the allocation above. A2, B and C10 all hold 5447 pairs, and the claim this paper turns on is the gap between C10 and B. Appendix B records what was held constant.

### 3.4 EXPERIMENTAL SETUP AND PRE-REGISTERED DECISION RULES

Every decision rule below was committed at a named commit before the runs it governs; amendments are append-only and dated; thresholds that failed are reported failed, and two that *passed* are reported and then refused.

**Endpoints and seeds.** The primary endpoint is COD10K-test  $S_\alpha$  from the final round’s best teacher checkpoint; NC4K  $S_\alpha$  is *reported, never decides*, as are MAE,  $F_\beta^w$  and  $E_\phi$ . Seeds are  $\{42, 43, 45\}$ . CHAMELEON is not an endpoint, for a reason that is itself a result of this paper (Section 5.2).

**The decision rule.** Three runs of the same arm do not return the same score, and the size of that seed-to-seed wobble is what any real difference has to beat. Let  $\hat{\sigma}$  be that wobble — the pooled within-arm standard deviation of  $S_\alpha$  across arms, at 8 degrees of freedom:

REAL EFFECT    iff  $\Delta > 2\hat{\sigma}$     and sign consistent 3/3 seeds  
 WITHIN NOISE    iff  $|\Delta| \leq 2\hat{\sigma}$   
 REAL REGRESSION    iff  $\Delta < -2\hat{\sigma}$     and sign consistent 3/3 seeds  
 INCONCLUSIVE    iff  $|\Delta| > 2\hat{\sigma}$  but sign not 3/3 — report as is, do *not* add seeds

So WITHIN NOISE means a gap smaller than the seed-to-seed spread, not a gap of zero. The rule is applied independently per architecture — SINet Fan et al. (2020) primary, SINet-v2 Fan et al. (2022) the robustness check, neither overriding the other, a disagreement reported as a disagreement. At  $n = 3$  we report no  $p$ -value, bootstrap or multiple-comparison correction: verdicts are  $\Delta$  against a  $2\hat{\sigma}$  bar with sign-consistency counts, and nothing converts them into significance claims.

**Two further arms that isolate direction from shape.** Arms C10 and B differ by *concentration* as well as by targeting, so that comparison alone cannot speak to the uncertainty signal. A separate campaign of 12 runs — two architectures  $\times$  two new arms  $\times$  three seeds, B and C10 re-scored rather than re-trained — holds the allocation’s shape fixed and varies only direction. **CSHUF** permutes the per-cluster scores across clusters, destroying targeting; **CINV** reverses their rank, making the allocation maximally anti-targeted. CSHUF’s permutation is the first of 64 candidate draws with zero fixed points and  $|\rho| \leq 0.10$ ; its realised rank correlation with the true score vector is  $-0.03351$ , against  $-1.0$  for CINV. Because equality is the *supporting* outcome here, set distinctness was gated before any GPU time at a Jaccard bound of 0.50: between 452 and 511 of each pair’s 1000 injected images differ.

Table 2: The verdict on COD10K-test  $S_\alpha$ , three seeds per arm.  $\hat{\sigma}$  is pooled within-arm at  $df = 8$ ; WITHIN NOISE means  $|\Delta| \leq 2\hat{\sigma}$ . The architectures disagree on the sign of the decisive gap. Per-run values: Table 6. *Source*: EXP ABC #3 @ REBUILD.LOG.txt:1376.

|          | $\hat{\sigma}$ (df = 8) | $2\hat{\sigma}$ | $\Delta(B - A2)$ | $\Delta(C - B)$     | Verdict      |
|----------|-------------------------|-----------------|------------------|---------------------|--------------|
| SINet    | 0.0090                  | 0.0179          | +0.0058 3/3      | + <b>0.0050</b> 3/3 | WITHIN NOISE |
| SINet-v2 | 0.0061                  | 0.0123          | +0.0030 3/3      | - <b>0.0004</b> 2/3 | WITHIN NOISE |

**Provenance of the rules.** Appendix B gives the commit carrying each rule and all three amendments, including one clause that no correct computation could satisfy.

## 4 RESULTS

Targeting did not beat random. Under the rule of Section 3.4, applied to 24 training runs,  $\Delta(C - B)$  is WITHIN NOISE on both architectures and both endpoints (Table 2), and the secondary endpoint returns the same verdict on every gap.

**The honest limit: this campaign resolved coarsely.** A verdict of WITHIN NOISE is only as strong as the noise, and here the noise was large enough that the bar a gap had to clear was *wider* than the effect the campaign was built to detect:  $2\hat{\sigma} = 0.0179$  against this repository’s reproduction gap of 0.0142, which it was declared able to half-resolve. At 126% of that effect, **this campaign could not have detected the reference effect had it been present**, and the null survives only on the tighter bar imported from the falsification campaign.

That bar is  $3.24\times$  tighter,  $\hat{\sigma} = 0.0028$  against 0.0090, pooled within-arm over  $\{B, C10, CSHUF, CINV\}$  at  $df = 8$ ; it applies here because that pool contains B and C10.  $\Delta(C10 - B) = 0.0050$  **against a bar of 0.0055 is  $0.91\times$  threshold, 3/3 sign-consistent: still WITHIN NOISE, and marginally so.** The asymmetry matters: where equality is the *supporting* outcome a tighter bar is conservative, but here the claim and the null coincide, so the looser bar is anti-conservative.

**The decisive gap is better resolved than either pooled bar implies.** Both bars are *pooled* across arms, and the looser is dominated by A0, whose seed spread here is 0.0173 — an arm not in the comparison at all. On the two arms actually compared the spread is 0.0018 (B) and 0.0041 (C10), and a Welch interval gives  $\Delta(C10 - B) = +0.0050$ , 95%  $[-0.0035, +0.0136]$  (Table 8). **That interval excludes the 0.0142 reference improvement** while admitting effects up to about two-thirds of it, so the honest statement is stronger than “we could not have seen it”. Two cautions, neither small: it is *post-hoc* and decides nothing, the frozen rule admitting no interval at  $n = 3$ ; and at  $df \approx 2.8$  the exclusion holds only just. A formal equivalence test at  $\delta = 0.005$  rejects nothing anywhere here — at three seeds that is a statement about the design, not the effect.

**The instrument is not blind.** A null is only worth reading if the same measurement can detect something real, so we checked that it can: we trained an arm we expected to be clearly worse and asked whether this design would see the difference. It did. A mean-teacher arm — SINet, three seeds, base pool, single-round by construction — reaches 0.6988 on COD10K-test against C10’s 0.7185:  $\Delta = +0.0197$  against a bar of 0.0060, or  $3.3\times$  the bar, sign-consistent 3/3, verdict DETECTED, the secondary endpoint agreeing.

This is a control on *instrument sensitivity*, not an ablation — the arm differs from C10 in method, round count and pool size at once, so no component attribution follows and none is made. **And it does not close the gap that matters most.** It shows effects of roughly the reference magnitude are visible; it does *not* show an effect the size of our nulls would be. The control’s own bar, 0.0060, exceeds the decisive gap  $\Delta(C10 - B) = 0.0050$ , so an effect that size would not have cleared this pooling either. A positive control at 0.020 licenses *the instrument works*, not *it would have seen the effect we are nulling*.

**Two loose ends, neither of which flatters us.** The campaign’s two largest effects sit just inside the bar, both at  $0.91 \times$  threshold — not a comfortable null, and a little more sensitivity would flip one to INCONCLUSIVE. And the ordering  $A0 < A2 < B \approx C10$  is monotone in all four cells, three steps sign-consistent 3/3: if real, padding the pool helps and targeting adds nothing. No step clears its bar, but we report it because it points away from our conclusion. Our C10 measures  $S_\alpha = 0.7185$  against the 0.7136 reported for the corresponding published configuration; Appendix K gives A0’s reproduction gate and spread.

The rest of this section reports the measurements that say *why*. Three of them were previously placed in the discussion; they are results, and they are reported as results. Each closes with what it transfers beyond this setup.

#### 4.1 CONCENTRATION RATHER THAN TARGETING

Suppose you aim a generation budget at the clusters where the model is least confident. The training set you get really does look different from a randomly chosen one of the same size: train a probe to tell the two apart and it succeeds easily, on images it has never seen. It is tempting to read that as evidence that the targeting worked. It is not.

We re-ran the selection with the uncertainty scores shuffled between clusters, so the budget was spread exactly as unevenly but pointed at the wrong places. The separation survived. We aimed at an arbitrary cluster instead: it survived again. Only removing the unevenness itself, by drawing two random sets of equal size, collapsed it (Figure 1a). The probe was detecting concentration, not the uncertainty signal.

The numbers follow the picture. Targeted selection separates from a random set of equal size at a held-out Cohen’s  $d$  — how far apart two sets are in standard deviations, scored on data the probe did not see — of roughly 1.00 to 1.23, an order of magnitude above a previously claimed 0.10. Permuting scores across clusters gives +0.9139 to +1.1403 in the same cells, and an *arbitrary* cluster as much again. Paired across twenty cells the signal adds +0.0073 of a  $d$  against its own shuffle, winning 13/20 ( $P \approx 0.13$ ) — weak evidence for a small positive contribution, not the coin flip we previously called it — and is *negative* at  $-0.0649$  against an arbitrary cluster, winning 4/20 ( $P \approx 0.006$ ). On the same yardstick the strongest directional evidence points *against* targeting. What targeting buys is narrowness — an effective-rank ratio of 0.634 over 20/20 cells — covering no more of the target manifold.

Geometry is not performance, so we trained it too. The falsification arms hold concentration *exactly* fixed — CSHUF permutes the per-cluster scores, CINV reverses their rank, both reproducing the committed allocation’s shape to five decimal places on all four shape statistics — and vary only direction. All twelve cells are WITHIN NOISE, the largest gap anywhere  $0.0050$  at  $1.82\hat{\sigma}$ . A directional pattern exists and we report it without claiming it: C10 is best in none of the four cells, the anti-targeted CINV has the highest mean in three of four, and the architectures disagree in sign on two of three gaps — the signature of noise.

**Scope, and the trained half is the weaker one.** Negligible *in this geometry* is supported; zero and harmful are not. The trained comparison is attenuated three times over and the factors multiply: injected images are 18.4% of the pool, the pinned budget dilutes exposure, and only  $\approx 48\%$  of injected images differ between arms — an effective contrast near 8.8% against a bar of 0.0055. That attenuation biases toward the null, which is why we disclose it; the next subsection removes the metric half of it. **Transfers:** the control that matters is not random selection but a same-shape allocation with the signal destroyed.

#### 4.2 THE CONCENTRATION EFFECT ON TRAINED ACCURACY

The trained comparison above is attenuated, and there is one place it should not be. The acquisition signal of Equation 1 is *boundary*-focused — its dominant term is an  $L_1$  distance between Sobel gradient magnitudes — while  $S_\alpha$  is structure-aware rather than boundary-specific. The campaign never measured the quantity the signal was built to move, so we re-scored all 36 runs’ committed predictions with Boundary IoU Cheng et al. (2021) and the contour  $F$ -measure (Table 10).

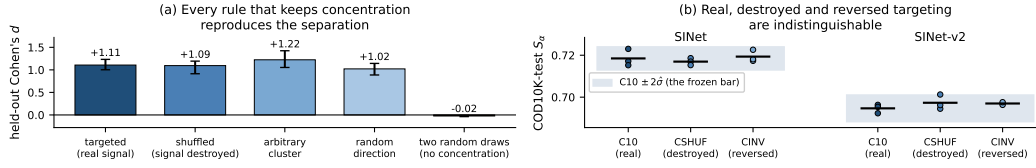


Figure 1: **(a)** How far a selected set sits from a random set of the same size, in held-out Cohen’s  $d$ . Every rule that keeps the allocation’s *concentration* reproduces the separation — permuting the uncertainty scores across clusters, and ignoring them entirely — and only destroying concentration collapses it. Bars are means over four cells, whiskers the range. **(b)** Trained  $S_\alpha$  for the three arms that hold concentration *exactly* fixed and vary only the signal’s direction; points are seeds, rules are arm means, the band is the  $\pm 2\sigma$  bar. *Source*: `c1_attribution.csv` (EXP C1 #4); `out/t2/abc_metrics.csv` (EXP T2 #3).

**On SINet the concentration effect becomes resolvable, and the reversed arm captures all of it.**  $\Delta(C10 - B)$  reaches  $+0.0138$  on Boundary IoU at  $1.37\times$  its bar and  $+0.0190$  on Boundary  $F$  at  $1.26\times$ , both 3/3 sign-consistent — REAL EFFECT under the same rule that returned WITHIN NOISE on  $S_\alpha$ . But  $\Delta(CINV - B)$ , the *anti-targeted* arm against the same baseline, is  $+0.0135$  and  $+0.0173$  — the same effect to within a tenth of a bar — while the direction contrasts  $\Delta(C10 - CSHUF)$  and  $\Delta(C10 - CINV)$  sit at  $0.37\times$  and  $0.04\times$ . A concentrated allocation trains a measurably better boundary than a dispersed one; *which* clusters it concentrates on does not. This is the paper’s thesis on accuracy rather than geometry, and the strongest form of it we have: the instrument that resolves concentration fails to resolve direction in the same cells, so the null on direction is not simply an absence of sensitivity.

**Three qualifications, all cutting against it.** These metrics were not pre-registered and re-decide no verdict; Table 2 stands. They binarise predictions where  $S_\alpha$  does not — though the pattern is not an artifact of where we binarise: swept over thresholds 0.4/0.5/0.6,  $\Delta(C10 - B)$  holds at  $1.39/1.37/1.23\times$  its bar and  $\Delta(CINV - B)$  at  $1.25/1.34/1.26\times$ , both 3/3 throughout, while both direction contrasts stay below  $0.52\times$  at every threshold. And **SINet-v2 resolves none of it** — every gap there is WITHIN NOISE — which we report rather than leading with the architecture that cooperates.

**Transfers:** a null on the endpoint metric does not license a null on the quantity the acquisition signal was designed to move. Measure that quantity, and run the same-shape control there too.

#### 4.3 CLUSTER STRUCTURE OF THE TARGET DISTRIBUTION

Allocating cluster by cluster presumes the data falls into clusters. Under every embedding we tested, it separates only weakly. The silhouette score — a standard measure of how cleanly points fall into groups, where values near zero mean the groups are not well separated — peaks at 0.1600 at  $k = 75$  in the space the pipeline uses, and the other two spaces we tried reach only 0.1465 and 0.0568, all below any conventional threshold, so the weakness is not an artifact of one encoder. The precise claim is *weak  $k$ -means separation under the tested embeddings*, not that the data has no structure: a partition can be poorly separated and still carry information. The third space fails more specifically: its silhouette falls monotonically from 0.0568 at  $k = 5$  to 0.0357 at  $k = 150$ , returning its smallest grid value by hitting the boundary rather than finding a maximum. Our guard required an interior maximum, and it failed. An arbitrary preprocessing choice reassigns 5.4% of memberships, so an effect smaller than that cannot be attributed to targeting.

**What the budget is actually aimed at.** A weakly separated partition is not yet a problem; spending a budget over it in a way that tracks something other than uncertainty is. The target pool is the two-dataset mixture of Section 3.1, whose components separate at up to 0.9225 AUC, and the allocation is not neutral between them. 51.5% of the budget lands on CAMO images, which are 24.8% of the pool: a  $2.08\times$  enrichment ( $2.14\times$  on the committed assignment, so not an artifact of re-deriving memberships at the 5.4% floor). The two largest quotas take 33.5% of the budget, are 90.2% CAMO, and hold 7 of the 2026 primary-endpoint images between them; weighted by budget,

the endpoint is represented at 0.72 of uniform density (Table 9, Figure 2). A third of the budget is aimed where the endpoint barely is, and “allocate by uncertainty” is in substantial part “allocate to CAMO”. Compounding this, only 20.5% of the endpoint’s  $1 - S_\alpha$  variance sits *between* clusters at all, so four fifths of what a targeted policy would have to fix is invisible to any cluster-level allocation however well aimed.

**Transfers:** three checks that cost no training should precede any cluster-wise allocation — the silhouette, the composition of the funded clusters against the pool they are drawn from, and the between-cluster share of endpoint error variance. A mixed target pool will quietly turn an uncertainty policy into a source-selection policy.

#### 4.4 UNCERTAINTY AND ERROR-TYPE ORDERING

An uncertainty score is only useful if it points at the errors you care about. In camouflage that means the boundary, which is what  $S_\alpha$  rewards and what a per-pixel average like MAE largely ignores. These signals track the per-pixel average better.

The ordering  $\rho(\text{MAE}) > \rho(1 - S_\alpha) > \rho(1 - \text{IoU})$  holds in **six independent signal–architecture tests**: three estimators — student–teacher disagreement, predictive entropy and ensemble disagreement — crossed with two architectures. The table carries eight rows because the ensemble is evaluated on two arms, but our own protocol treats the second as a *sensitivity check* on the first, so they are not independent and we do not count them twice. The gap spans +0.0708 to +0.3330, all clearing the resolution floor of about 0.06, the smallest only just. The same criterion can fail, and elsewhere it does: aggregated over a boundary band, only 2 of 8 rows pass and four go negative, outside our declared interpretation space, which did not anticipate inversion.

**The area control.** A whole-image mean is uncertainty density times area, so if MAE and the signals both scale with area while  $S_\alpha$  carries an object-size term that does not, the ordering could be an artifact of the metrics. We partialled area out (Table 5). **The evidence against the confound is the direction of the shift, not the pass count: partialling object area raises  $\rho(1 - S_\alpha)$  in all eight rows**, the opposite of what the confound predicts. The ordering survives on 7/8 and 6/8 rows against a floor of 6/8 declared before the run — a weak floor,  $P \approx 0.145$  under a fair coin, so the uncertain-area covariate clears a near-chance bar by zero margin. **Area accounts for part of the gap but not its sign.**

**Scope:** correlational, one endpoint, and the *magnitude* does not generalise —  $\rho(1 - S_\alpha)$  reaches +0.5628 on SINet-v2, so the claim is strictly ordinal. **Transfers:** if you care about structural quality, uncertainty of these kinds is a better proxy for pixel-average error than for what you want to improve.

## 5 DISCUSSION

Two things are established above and should not be run together. The **geometric** claim is positive and well resolved: concentration, not the uncertainty direction, is what makes a targeted set different from a random one, and the same-shape shuffle is the control that shows it. The **downstream** claim is a null under stated sensitivity: on trained accuracy the direction of the signal moves nothing we can resolve, and on the one family of metrics where the concentration effect *is* resolvable, the reversed arm captures it in full. A reader who takes the second claim as *proof* that uncertainty guidance cannot work has read more than we measured; a reader who takes the first as merely suggestive has read less. What remains is to say what the null is scoped by, and to give a second case of the same methodological point.

### 5.1 SETUP-SPECIFIC LIMITATIONS

The causes above would apply to anyone. Three more are ours alone. The optimisation budget is pinned at 253 and 127 steps per epoch regardless of pool size, so Stage C’s 1000 images *displace* base-pool exposure rather than adding to it — each image is seen 0.74 times per epoch instead of 0.91. The foreground supply is exhausted: the render set is a bijection onto the base set and not one traced object shows any sign of regeneration, so objects are copied rather than synthesised. And the

generated distribution is narrower than its own input — real photographs of a different genre cover the target manifold better than synthetic images generated from those very photographs.

Each has an obvious rebuttal and each rebuttal is right: the first is a property of the training schedule, not of synthetic data; the second follows from choosing a generator that composites by design; the third is a distributional characterisation, so “the distribution is far, therefore no amount of it can help” would be a non-sequitur. **The first is also the largest open question this paper creates**, and it is not symmetric between the arms: B and C10 hold identical pool sizes and therefore receive identical step counts under either schedule, so the pin cannot bias the decisive gap — what it changes is whether the added images are trained on or merely substituted in. None of the three licenses the conclusion that closed-loop generation cannot work; they say only that this build could not have shown that it does. We also *withdraw* one of our own candidate causes, by source reading rather than measurement. Appendix E gives every measurement, the withdrawn cause, and the evidence that the coverage result does not depend on statistical power.

## 5.2 A SECOND CASE: A CONTROL THAT COULD NOT HAVE RETURNED ANYTHING ELSE

One methodological point runs through this paper: *a control is informative only if it could have returned something else*. Section 4.1 makes it about an experimental arm — comparing against random selection could not have distinguished aiming from concentrating, because random selection differs from targeting in two ways at once, and only the same-shape shuffle separates them. This subsection makes the same point about an *evaluation*, on a finding that stands entirely independently of whether our method worked.

CHAMELEON is one of the four benchmarks camouflage papers report on, and most of it is training data. The usual check hashes the image files and looks for identical bytes; run it here and it reports no overlap at all. It is not wrong — it answers a different question. The copies were re-saved in a different format, so not one byte matches while every pixel does. A check that returns 0 on a benchmark that is two-thirds training data is the same failure as a control that returns a separation whatever you aim it at: in both cases the instrument cannot fail, so its passing carries no information.

**The finding, and the control that misses it.** We report two tiers. The floor is what a released same-dimension detector reproduces on any machine; the extension adds copies that were rescaled or cropped. **Floor:** 41 of CHAMELEON’s 76 images — 53.9% — are the same photographs as images in the COD10K-train and CAMO training pool, re-encoded, 40 in the public COD10K-train split and one in CAMO. A model trained on COD10K-train has already seen over half of CHAMELEON before being evaluated on it. We verified against an author-sourced release, byte-identical at 76/76. Exact hashing returns 0 collisions and is correct at that level: these are re-encodings, not byte-identical copies. The number was right; the inference drawn from it was not — and we hit the same trap ourselves, our first implementation reporting 10/76 (13.2%) because contrast normalisation destroyed the discriminative scale. CHAMELEON predates COD10K, so the parsimonious reading is that COD10K-train absorbed these images, though only pool membership is measured.

**41 was itself a lower bound, and we can now say by how much.** Our detector compares only same-dimension images, so a copy that was *rescaled* or *cropped* out of its dimension group is invisible to it and reported UNCHECKED, not clean — 25 of the 76. SIFT correspondences filtered by a RANSAC homography close most of that gap. (We also tried resolution-normalised global matching; it recovered nothing these keypoints did not, and we report it only so its null is on the record.) The inlier threshold is calibrated on answers already known rather than chosen by inspection: 525–1174 inliers on the 41 confirmed pairs against a maximum of 8 on random pairs and 19 anywhere in the NC4K control, and we take the operating point one inlier above that worst case, at 20.

An inlier count, however, only shows that local patches agree on a geometric model — two exposures of one static scene agree on a homography too. Since the claim is that these are the same photographs, we measure it as one: each candidate partner is warped into the CHAMELEON frame through the recovered homography and we take the residual over the overlap. Run first on the 41 known pairs the instrument reproduces their recorded mean absolute difference to three decimals, which is what makes its verdict on the rest readable. **Nine of the 25 survive:** residual 3.8 to 12.1 on a 0–255 scale, correlation 0.96 to 1.00, shear and perspective indistinguishable from zero, at scales from  $0.39\times$  to  $6.36\times$  — and two of the nine at scale 1.000 with different dimensions, which is to

say plain crops. A tenth candidate cleared the inlier threshold and *failed* this test: its homography decomposes to  $-86.5^\circ$  of shear and anisotropic scale 0.24 against 1.24, collapsing the frame towards a line, and at full resolution it yields no homography at all. We exclude it. **CHAMELEON is therefore 50/76 (65.8%) training data** — 44 of the partners in the public COD10K-train split and 6 in CAMO — **with 16 unresolved and 10 clean with respect to the training pool** (Figure 4, calibration in Appendix H). Every one of the nine left its dimension group by a rescale or a crop, which is why the released detector could not see it. The 16 are where the count can still move: each lacks a partner at either tier, and UNCHECKED is not a synonym for clean. 50 is therefore a floor — the number can rise with a better instrument, and cannot fall without overturning a pixel measurement.

**A caution for anyone repeating this across benchmarks.** Exact hashing returns 0 against the *training pool*, which is the claim we make. It does not return 0 across benchmarks: three CHAMELEON images are byte-identical to CAMO-250 images (*animal-17*, *animal-25*, *animal-60*). Those files are not in the training pool, so they do not enter the count above, but CAMO-250 is the split this repository selects checkpoints on — so a further three test images sit in the selection path, and one of them is an image we otherwise call clean.

**The count is a property of the data, not of a tolerance.** Sweeping the tolerance saturates the count at 41, and the sorted nearest-neighbour distances independently place 41 images below 5.51 with the next at 40.58 — a  $7.36\times$  jump, unchanged at every shortlist depth. Two instruments that owe each other nothing land on the same integer, and the other three endpoints show no jump.

**A negative control, from the same instrument.** An audit that finds contamination wherever it looks is worthless. The identical detector returns a clean null on NC4K against the COD10K test split — 0/4121 at every tolerance swept, minimum nearest distance  $3.44\times$  the confirmation tolerance, and 0 of the 1082 images actually comparable — and so does the geometric extension, at 0 of 120 sampled, its highest inlier count anywhere being 19 against an operating point of 20. Appendix H gives both, and the reason NC4K’s 1/4121 in Table 7 is a different axis that must not be collapsed with it.

**Non-independence, not inflation.** We expected to say how much a contaminated column flattens a score, measured it, and found nothing: all eight difficulty percentiles for the leaked subset land within 0.448 to 0.559, and the sign of the clean-minus-leaked difference *flips* with the mask release. The pre-declared reading — a percentile inside  $[0.25, 0.75]$  means no skew — is what stopped us writing the quotable sentence. Our claim is therefore non-independence, not inflation: *CHAMELEON is not an independent endpoint for COD10K-trained models*.

**The protocol: retire the benchmark, do not subset it.** Our released protocol originally recommended reporting CHAMELEON on its 35 uncontaminated images. We withdraw that: of 76 images, 50 are now confirmed contaminated and 16 more are *unchecked, not clean*, leaving 10 **clean with respect to the training pool** — one of which, *animal-25*, is byte-identical to a CAMO-250 image and so is not independent of that benchmark either. The extension makes the recommendation stronger rather than weaker — every image it moved came out of the unchecked column, and none moved the other way. A 10-image endpoint is not an endpoint, and with two incompatible mask releases the honest recommendation — a change to one we already released — is to retire CHAMELEON rather than subset it. Report COD10K-test and NC4K instead, disclosing the unchecked share. The detector we release is checkable without our data, and Appendix H records its self-test, the extension’s calibration and the per-pair adjudication.

**A concrete instance, from the protocol we test.** The method under test evaluates on CHAMELEON, and its task setup states that for this family of benchmarks it “*consistently use[s] the real unlabeled COD10K training set as the target domain*”. It takes a precaution: when the test set is CHAMELEON, those images are removed from the *source* domain “*to prevent target domain data leakage*”. That closes the source path but not the target pool, since the case definitions adjust source selection only — and that pool holds 44 of the 76 CHAMELEON images, so a model following this protocol self-trains on pseudo-labels over a set containing more than half the images it

is then tested on. We attribute no intent: the source-side removal shows the authors were alert to leakage, and the target-side path is invisible without a re-encode-tolerant check.

## 6 CONCLUSION

We built a loop that aims a synthetic-data budget by the model’s own uncertainty, and no advantage survived measurement: the decisive gap sits inside the seed spread, and an interval on the two arms compared excludes the improvement reported for this approach while admitting effects up to about two-thirds of it. A positive control confirms the design detects a real effect of  $+0.0197$ , though not one the size of the null.

Two kinds of evidence sit underneath that, and the paper is worth reading only if they are kept apart. The *geometric* result is positive and we state it without hedging: concentration, not the uncertainty direction, produces the separation that makes targeted selection look like it works, and the control that shows this is a same-shape shuffle rather than a random baseline. Where that concentration effect becomes resolvable on trained accuracy — boundary metrics on one of two architectures — the *anti-targeted* arm captures it in full. The *downstream* result is a null under stated sensitivity and nothing more: it does not show that uncertainty guidance cannot work, and we have named the sensitivity at which it would have shown one.

The idea still seems reasonable. For it to have a chance you would need an optimisation budget that grows with the pool rather than one under which added images displace real ones, a foreground supply not already exhausted, a generator wider than its own input, and a target pool that is not a mixture whose components the uncertainty signal can tell apart — ours sends half its budget to a quarter of its data. Separately: CHAMELEON is 65.8% training data for models trained on COD10K — a floor, with 16 of its 76 images still unresolved — the usual check reports it perfectly clean, and we release one that does not. A control that returns 0 is only as good as its ability to return something else.

## AI USE STATEMENT

Generative AI tools were used in the preparation of this work as coding and writing assistants: for drafting and refactoring analysis and experiment-driver code, for drafting and editing prose, and for literature search. They were not used to generate, select, alter or interpret any experimental result, and no number in this paper originates from a language model.

The design decisions that determine every reported verdict — the arms, the endpoints, the seeds, the noise estimate and the four verdict bands — are recorded in pre-registration documents committed before the runs they govern (Section 3.4), and amendments to them are append-only and dated. Verification does not rest on trusting the tools or the authors: **every numeric value in this paper is traceable to a log block emitted by a committed script, and no result was transcribed by hand.** Appendix C records that mapping, and the released artifacts allow any reported number to be regenerated from primary data. AI-assisted code was reviewed by the authors and is itself committed, so it can be audited rather than taken on trust; where an AI-assisted analysis produced a result that the pre-registered rule did not sanction, it is reported as post-hoc and marked as deciding nothing (Tables 8 and 10).

The authors take full responsibility for the content of this work, including any part drafted with AI assistance.

## ETHICS STATEMENT

This work involves no human subjects and no personally identifying data. It uses four public research benchmarks for camouflaged object detection — COD10K Fan et al. (2022), CAMO Le et al. (2019), NC4K Lv et al. (2021) and CHAMELEON Skurowski et al. (2018) — together with the HKU-IS salient-object set as a source pool and the released LAKE-RED Zhao et al. (2024) generator checkpoint. Each is used for non-commercial academic research, which is the use its distribution terms provide for, and each is obtained from its authors’ own release rather than redistributed by us; we release no benchmark imagery, only the derived pair list and detector described below.

CHAMELEON in particular is an unpublished collection that has become a de facto standard, which is part of why its provenance is worth auditing at all.

Section 5.2 reports an overlap between two public datasets. We state explicitly what that measurement does and does not support. It is a measurement of pool membership; it attributes no intent to any dataset’s authors, and we draw the direction of provenance as an inference from release chronology rather than as a finding. We measured for a score-inflation figure, did not find one, and do not report one. The one correction we direct at anyone is directed at ourselves: it was our own protocol that adopted CHAMELEON as an evaluation endpoint without checking whether the repository we built on ever sanctioned it. We release the detector so that the measurement can be repeated, contested, or extended by others rather than taken on trust.

## REPRODUCIBILITY STATEMENT

The decision rules governing every result are committed before the runs they govern, at the commits named in Section 3.4, and amendments to them are append-only and dated. Each experiment has a committed result document and one or more committed log blocks; each number in this paper is traceable to such a block or to a committed artifact, and Appendix C gives that mapping. Per-run metrics for all 36 training runs, the full clustering sweep, and the full signal-generalizability table are in the appendix. The contamination detector is released with an eight-assertion known-answer self-test that runs without any of our data, alongside the confirmed pair list, the clean protocol, and the calibration of the geometric extension.

**Artifacts.** Code, pre-registration documents, per-run metrics, the append-only experiment log, and the released detector accompany this submission as supplementary material, which preserves anonymity without a third-party host. The supplement is self-contained: every experiment is one directory holding its scripts, its declared thresholds and its outputs, and each cites the log block that produced it.

Two limits on reproducibility. Our regeneration check establishes byte-identical reproduction on one machine with one software stack; it is not a claim of determinism across hardware or driver versions. And a render from this generator is a function of its foreground, its mask, *and its position within the processing shard*, so changing the shard count changes the outputs even at a fixed seed.

## A ADDITIONAL RELATED WORK

**COD benchmarks and detectors.** The standard COD evaluation suite comprises COD10K Fan et al. (2022), a large benchmark with ten camouflage categories and pixel-accurate masks; CAMO Le et al. (2019), which focuses on man-made camouflage and occlusion; NC4K Lv et al. (2021), assembled from Flickr; and CHAMELEON Skurowski et al. (2018), an earlier unpublished collection that has become a de facto standard. The continued use of all four matters for Section 5.2: an image shared by a training set and any reported test set contaminates evaluation for methods using the full suite. SINet Fan et al. (2020) decomposes COD into search and identification branches, while SINet-v2 Fan et al. (2022) adds a neighbour connection decoder for boundary refinement; we use both off-the-shelf. SegMaR Jia et al. (2022) iteratively magnifies and re-processes ambiguous regions, representing architectural progress outside our allocation experiment. We report the structure-aware S-measure Fan et al. (2017), enhanced-alignment E-measure Fan et al. (2018), and weighted  $F_{\beta}^w$  Margolin et al. (2014) for comparability with S2R-COD.

**Synthetic generation and domain adaptation.** Latent diffusion Rombach et al. (2022) and DDIM Song et al. (2021) provide the high-resolution generator and deterministic sampling schedule inherited by our pipeline. Copy-paste augmentation Ghiasi et al. (2021) similarly composes object cut-outs with varied backgrounds; LAKE-RED applies this compositional idea while retrieving background patches. CyCADA Hoffman et al. (2018), in contrast, aligns source and target feature distributions with cycle-consistent adversarial training; our loop does not operate at the feature level and instead feeds generator output into a self-training objective. Other camouflage-image generators exist Chen et al. (2025), but they are outside the experiment, so the reported null is bounded to LAKE-RED.

**Uncertainty-guided acquisition and self-training.** FixMatch Sohn et al. (2020) couples confidence thresholding with consistency regularisation; our pseudo-labels likewise arise in an inter-round consistency step, but drive a second training round rather than an online update. Pseudo-labelling Lee (2013) supplies the hard-target view of this step. DINOv2 Oquab et al. (2024) and CLIP Radford et al. (2021) provide the frozen representations used for clustering. These choices complement core-set selection Sener & Savarese (2018): the clusters spread the budget, while the disagreement scores determine which clusters receive more of it. BADGE Ash et al. (2020) jointly uses uncertainty and diversity in gradient-embedding space; our same-shape shuffle in Section 4.1 is a direct control for the corresponding entanglement in this allocation.

**Benchmark integrity and negative results.** Dataset-collection procedures can induce shared statistical bias Torralba & Efros (2011), and near-duplicate analyses of CIFAR and ImageNet show that visually identical images can cross split boundaries Barz & Denzler (2020). Related contamination concerns appear in language-model evaluation Wu et al. (2024); the mechanism differs here, but the need for explicit auditing is shared. Our pre-registration follows reproducibility guidance Pineau et al. (2021) and the case for publishing disconfirming results Karl et al. (2024). Finally, criticisms of learned comparison measures under resizing and representation choices Parmar et al. (2022); Jayasumana et al. (2024) motivate treating encoding-sensitive probes as diagnostics rather than as clean perceptual evidence.

## B DETAILED METHOD SPECIFICATION

**Training.** SInet Fan et al. (2020) at batch 16 for 40 epochs and SInet-v2 Fan et al. (2022) at batch 32 for 100 epochs, Adam at  $10^{-4}$ , images at  $352 \times 352$  with ImageNet normalisation. The EMA teacher uses momentum 0.996 under this task setting. The consistency weights in Equation 1 are the configured  $a = 0.9$ ,  $b = 0.3$  with the unweighted cross-entropy variant — note that these are the values the trainer sets for this task, not the loss class’s own defaults. Strong views add random autocontrast and Gaussian blur to the weak view.  `cudnn.deterministic`  is enabled and  `cudnn.benchmark`  disabled; this reduces, and is not claimed to eliminate, kernel nondeterminism.

**What was held constant, verified.** Across all 24 runs the base pool is asserted byte-identical against a hash manifest with 0 mismatches, both rounds read the unfiltered 4040-image target set, and every run shares one trainer invocation, checkpoint rule, endpoint set and deterministic-kernel setting.

**Optimisation budget.** Steps per epoch are set by the shorter of the source and target loaders, which fixes them at 253 (SInet) and 127 (SInet-v2) in both rounds of every run regardless of pool size. This is measured per run and asserted, and it is the mechanism behind Section 5.1’s first cause.

**Allocation sweeps.** Stage (iv) matches against a grey-masked cutout rather than a render because at selection time the foreground exists and the render does not. The budget grid is  $B \in \{250, 500, 1000, 2000, 3000, 4447\}$  and the temperature grid over  $\alpha$  is extended until both degenerate ends are reached, defined numerically as concentrated when  $\max_j p_j \geq 0.95$  and uniform when the total variation from uniform falls below 0.01, with caps at  $\alpha \in [10^{-4}, 10^3]$ . The campaign uses  $\alpha = 1.0$ ,  $B = 1000$ .

**Metrics.**  $S_\alpha$  at  $\alpha = 0.5$  is the primary endpoint metric; MAE is the checkpoint-selection metric;  $F_\beta^w$  and  $E_\phi$  are computed and reported but decide nothing. The  $F$ -measures apply  $\beta$  in the non-squared form, so the metric name rather than a formula is what identifies them.

**The A2 control’s one confound.** A2 pairs the authors’ original photographs with the authors’ masks, whose mean foreground fraction differs from the raw masks by 0.0058 on 1000 of 5447 images. This affects  $\Delta(B - A2)$  and is disclosed; it does not touch  $\Delta(C10 - B)$ , the decisive gap, since B and C10 share mask provenance, pool size and render pool.

**Pre-registration commits and amendments.** The four-arm rule is recorded at 065dac6, the commit carrying that campaign’s first log block; the falsification campaign’s at 617a2e1, before its first training run; the signal-generality check’s at 6efde9f; the area control’s at c2114af; the positive control’s at c2114af. Three amendments exist, all append-only and dated. Two are reference-tolerance defects of the same kind: a frozen clause requiring reproduction to better than  $10^{-9}$  against a file recording six decimal places, and — in the area control, written by the same author who had just documented the first — the identical mistake against a file recording four. Both are replaced by exact agreement at the recorded precision, which is the strictest satisfiable form, and neither altered a decision threshold. The third marks one verification gate not applicable in scope after it was found to compare two architectures’ signals.

## C SUPPLEMENTARY RESULTS AND TABLES

**The area control in detail.** The covariate is object area on the endpoint side and uncertain-region area on the target side. Partialling object area raises  $\rho(1 - S_\alpha)$  in all eight rows — +0.3433 to +0.4988 for student–teacher disagreement on SINet. The single object-area failure is a tie crossing by 0.0012, and both uncertain-area failures are on SINet-v2, where  $\rho(\text{MAE})$  falls +0.7070 to +0.4457 once band area is removed.

Table 3: Full clustering sweep over the target set: silhouette (mean over 10  $k$ -means seeds), bootstrap ARI and seed ARI, for every  $k$  in the declared grid and all three embedding spaces. The principled  $k$  per space is **bold**. No  $k$  in any space reaches a silhouette conventionally regarded as well-separated structure, and clipL224 falls monotonically, so its selection criterion returns the grid edge rather than an interior maximum — recorded as a FAIL of the declared guard. *Source:* rebuild/B1/out/b1\_k\_sweep\_{dinoL224,dinoL518,clipL224}.csv; block EXP B1 #3 @ results/REBUILD-LOG.txt:922.

|     | dinoL224      |        |        | dinoL518      |        |        | clipL224      |        |        |
|-----|---------------|--------|--------|---------------|--------|--------|---------------|--------|--------|
| $k$ | sil.          | bs.ARI | sd.ARI | sil.          | bs.ARI | sd.ARI | sil.          | bs.ARI | sd.ARI |
| 5   | 0.0538        | 0.712  | 0.795  | 0.0566        | 0.624  | 0.732  | <b>0.0568</b> | 0.953  | 0.983  |
| 10  | 0.0868        | 0.676  | 0.709  | 0.0923        | 0.598  | 0.770  | 0.0550        | 0.692  | 0.805  |
| 15  | 0.1053        | 0.608  | 0.714  | 0.1152        | 0.573  | 0.592  | 0.0486        | 0.575  | 0.670  |
| 20  | 0.1180        | 0.629  | 0.693  | 0.1326        | 0.506  | 0.524  | 0.0486        | 0.635  | 0.646  |
| 30  | 0.1378        | 0.596  | 0.642  | 0.1527        | 0.539  | 0.617  | 0.0488        | 0.609  | 0.631  |
| 50  | <b>0.1465</b> | 0.610  | 0.615  | 0.1556        | 0.579  | 0.600  | 0.0448        | 0.498  | 0.549  |
| 75  | 0.1457        | 0.588  | 0.627  | <b>0.1600</b> | 0.579  | 0.616  | 0.0396        | 0.463  | 0.477  |
| 100 | 0.1419        | 0.608  | 0.637  | 0.1557        | 0.583  | 0.601  | 0.0382        | 0.419  | 0.452  |
| 150 | 0.1367        | 0.630  | 0.636  | 0.1507        | 0.618  | 0.625  | 0.0357        | 0.380  | 0.409  |

## D POST-HOC DIAGNOSTICS: INTERVALS, ALLOCATION COMPOSITION, BOUNDARY METRICS

Everything in this appendix is **post-hoc and decides nothing**. The verdicts of Section 4 rest on the pre-registered  $2\hat{\sigma}$  rule, which admits no  $p$ -value, bootstrap or interval at  $n = 3$ ; these analyses were run after those verdicts were committed, on the same committed artifacts, and are reported because a reader is entitled to ask what interval the data is consistent with even where the frozen rule declines to answer. They are generated by rebuild/DIAG/ and consolidated in diagnostics.summary.json.

**On reading the equivalence tests.** No comparison in this paper establishes equivalence at  $\delta = 0.005$ , and at three seeds per arm none could: the smallest  $\delta$  the decisive gap could have established is 0.0113. We report this rather than the  $p$ -values because the  $p$ -values would invite the reading that equivalence was tested and failed, when what failed is the design’s power to test it. The intervals in the same table are the informative part, and Section 4 states what they support.

Table 4: Full signal-generality measurement, all 16 rows: four signal variants across three signal families, two architectures, two aggregations.  $\rho$  at the committed seed,  $\pm$  sd over 10  $k$ -means seeds. The whole-image block passes the pre-registered ordering 8/8; the boundary block passes 2/8 with four rows going negative, an outcome outside the pre-registered interpretation space, reported as it fell. *Source:* rebuild/T2C/out/t2c\_table.csv; block EXP T2C @ results/REBUILD\_LOG.txt:2550.

| arch     | signal         | agg.  | $\rho(\text{MAE})$   | $\rho(1 - S_\alpha)$ | $\rho(1 - \text{IoU})$ | order | seeds |
|----------|----------------|-------|----------------------|----------------------|------------------------|-------|-------|
| SINet-v2 | ES             | whole | $+0.7070 \pm 0.0404$ | $+0.5628 \pm 0.0359$ | $+0.4591 \pm 0.0315$   | PASS  | 10/10 |
| SINet-v2 | ensemble A0    | whole | $+0.6191 \pm 0.0675$ | $+0.5483 \pm 0.0409$ | $+0.4540 \pm 0.0451$   | PASS  | 9/10  |
| SINet-v2 | ensemble CSHUF | whole | $+0.5576 \pm 0.0690$ | $+0.4488 \pm 0.0580$ | $+0.3370 \pm 0.0577$   | PASS  | 10/10 |
| SINet-v2 | entropy        | whole | $+0.7393 \pm 0.0309$ | $+0.5404 \pm 0.0408$ | $+0.4070 \pm 0.0395$   | PASS  | 10/10 |
| SINet-v2 | ES             | bound | $-0.5485 \pm 0.0745$ | $-0.1146 \pm 0.0675$ | $+0.0537 \pm 0.0558$   | FAIL  | 0/10  |
| SINet-v2 | ensemble A0    | bound | $-0.1960 \pm 0.0524$ | $+0.0425 \pm 0.0410$ | $+0.0933 \pm 0.0396$   | FAIL  | 0/10  |
| SINet-v2 | ensemble CSHUF | bound | $-0.2527 \pm 0.0668$ | $-0.0591 \pm 0.0609$ | $-0.0011 \pm 0.0579$   | FAIL  | 0/10  |
| SINet-v2 | entropy        | bound | $+0.4208 \pm 0.0356$ | $+0.4182 \pm 0.0455$ | $+0.3605 \pm 0.0470$   | PASS  | 8/10  |
| SINet    | ES             | whole | $+0.6284 \pm 0.0381$ | $+0.3433 \pm 0.0301$ | $+0.2613 \pm 0.0383$   | PASS  | 10/10 |
| SINet    | ensemble A0    | whole | $+0.6028 \pm 0.0466$ | $+0.2698 \pm 0.0487$ | $+0.1491 \pm 0.0560$   | PASS  | 10/10 |
| SINet    | ensemble CSHUF | whole | $+0.5212 \pm 0.0564$ | $+0.4006 \pm 0.0459$ | $+0.3361 \pm 0.0447$   | PASS  | 10/10 |
| SINet    | entropy        | whole | $+0.5892 \pm 0.0368$ | $+0.4267 \pm 0.0277$ | $+0.3588 \pm 0.0351$   | PASS  | 10/10 |
| SINet    | ES             | bound | $-0.0012 \pm 0.0481$ | $-0.0567 \pm 0.0500$ | $-0.0554 \pm 0.0479$   | FAIL  | 8/10  |
| SINet    | ensemble A0    | bound | $+0.3633 \pm 0.0605$ | $+0.2910 \pm 0.0509$ | $+0.2215 \pm 0.0513$   | PASS  | 10/10 |
| SINet    | ensemble CSHUF | bound | $+0.3175 \pm 0.0637$ | $+0.3857 \pm 0.0381$ | $+0.3770 \pm 0.0322$   | FAIL  | 2/10  |
| SINet    | entropy        | bound | $+0.2434 \pm 0.0414$ | $+0.2492 \pm 0.0540$ | $+0.2432 \pm 0.0563$   | FAIL  | 6/10  |

Table 5: The area control (EXP AC). First-order partial Spearman on the same cluster partition and floors as the committed measurement.  $z_{\text{obj}}$  = per-cluster mean object area on the endpoint side;  $z_{\text{unc}}$  = per-cluster mean uncertain-region area on the target side. Ordering survives 7/8 and 6/8 against a floor of 6/8 declared before the run. Note that partialling object area *raises*  $\rho(1 - S_\alpha)$  in every row — the opposite of the confound’s prediction. *Source:* rebuild/AC/out/ac\_table.csv; block EXP AC @ results/REBUILD\_LOG.txt:2612.

| arch           | signal         | raw     |                | $ z_{\text{obj}} $ |                | $ z_{\text{unc}} $ |                |
|----------------|----------------|---------|----------------|--------------------|----------------|--------------------|----------------|
|                |                | MAE     | $1 - S_\alpha$ | MAE                | $1 - S_\alpha$ | MAE                | $1 - S_\alpha$ |
| SINet          | ES             | +0.6284 | +0.3433        | +0.5761            | +0.4988        | +0.5518            | +0.3897        |
| SINet          | entropy        | +0.5892 | +0.4267        | +0.5505            | +0.5517        | +0.5060            | +0.4426        |
| SINet          | ensemble A0    | +0.6028 | +0.2698        | +0.5411            | +0.4301        | +0.5603            | +0.2583        |
| SINet          | ensemble CSHUF | +0.5212 | +0.4006        | +0.5002            | +0.4857        | +0.4857            | +0.3937        |
| SINet-v2       | ES             | +0.7070 | +0.5628        | +0.7062            | +0.6272        | +0.4457            | +0.4744        |
| SINet-v2       | entropy        | +0.7393 | +0.5404        | +0.7134            | +0.6546        | +0.5369            | +0.4266        |
| SINet-v2       | ensemble A0    | +0.6191 | +0.5483        | +0.6245            | +0.5969        | +0.4355            | +0.4488        |
| SINet-v2       | ensemble CSHUF | +0.5576 | +0.4488        | +0.5344            | +0.5209        | +0.4482            | +0.3615        |
| ordering holds |                | 8/8     |                | 7/8                |                | 6/8                |                |

**Subgroup breakdowns.** Splitting the endpoint into tertiles by object size, by endpoint uncertainty and by camouflage difficulty does not change the picture of Section 4.2: on SINet the C10 – B boundary-IoU gap clears its bar in 8 of 9 cells, and on SINet-v2 in 0 of 9. No slice shows a targeting effect that the whole set hides. These are exploratory at three seeds and no verdict is claimed from them.

## E SETUP-SPECIFIC LIMITATIONS AND DIAGNOSTICS

These causes are properties of this loop and this generator. They explain why this attempt could not have demonstrated a gain; they do not license the conclusion that closed-loop generation cannot work. Each carries a legitimate dismissal, which we state.

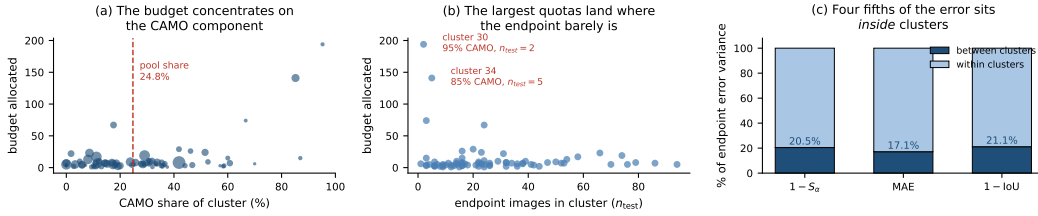


Figure 2: **(a)** Each point is one of the 75 clusters, sized by how many target images it holds. The budget piles onto clusters far to the right of the pool’s own CAMO share, so the allocation is partly a source selector. **(b)** The same budget against how many primary-endpoint images each cluster contains: the two largest quotas sit at  $n_{test} = 2$  and 5. **(c)** Endpoint error variance decomposed across the same clusters — four fifths of it is within clusters, so a cluster-level policy cannot address it however well aimed. *Source:* rebuild/DIAG/out/diag\_clusters.json.

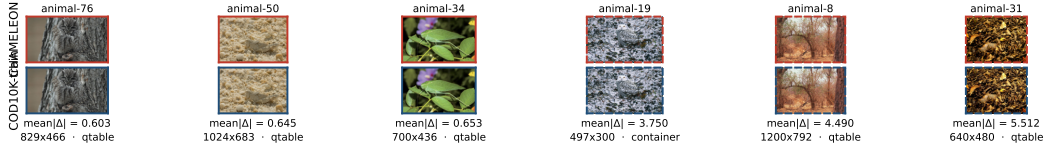


Figure 3: Six of the 41 confirmed pairs: the author-sourced CHAMELEON image (top) and its partner in the COD10K-train pool (bottom). *Selection rule, applied not curated:* the six lowest mean absolute difference among pairs with a COD10K-train partner. Exact hashing reports 0 collisions for every pair shown, each being a re-encoding rather than a byte-identical copy. *Source:* chameleon\_contaminated.json (EXP D2R #4).

*The optimisation budget is pinned at 253 and 127 steps per epoch for the two architectures, in both rounds of every run, regardless of pool size — so adding 1000 images to a 4447-image pool bought no additional optimisation, only a change of mixture. A reviewer may call this a property of the training schedule rather than a result about synthetic data, and that reading is correct: nothing here measures what happens when the optimisation budget grows with the data, which is the largest open question this paper creates.*

*The foreground pool is exhausted; the background is not.* The render set is a bijection onto the base foreground set — all 4447 renders trace into it, 0 outside, 0 base foregrounds unrendered — on the pool training reads as well as on ours, and the mapping is literal: interior error against the source foreground is 5.603 and 9.633 mean absolute grey levels against background error near 70.6. Of the 8885 objects with a valid interior comparison (4443 + 4442, since erosion leaves no interior at all for four and five objects each under 1% of frame), 0 show any sign of regeneration. This is an identity statement about pixels, not a claim about similarity, which is what makes it resistant to the obvious counter-argument. The dismissal — this generator composites by design, so of course it adds no new foregrounds — is right, and what survives it is the scope statement every null here inherits: evidence about targeting under an exhausted foreground supply, silent on whether a larger or more diverse one would behave differently.

*The generated distribution is narrow, and this evidence alone does not depend on power.* As recall against the target manifold the synthetic pools reach 0.1277 to 0.5421 across three embedding spaces, while the generator’s own input pool — real photographs of a different genre — reaches 0.7097 to 0.7473 and a different real camouflage dataset reaches 0.7921 to 0.8995: real photographs of the wrong genre cover the target better than synthetic images generated from those very photographs, a paired loss of  $-23.6\%$  to  $-82.9\%$ . Effective rank falls 243.3 to 193.2 while precision rises 0.6490 to 0.6886 — narrowing, not displacement. This is a distributional characterisation, not a training-utility bound; reading it as “the distribution is far, therefore no amount of it can help” would be a non-sequitur, and what generalises is the design, not the number.

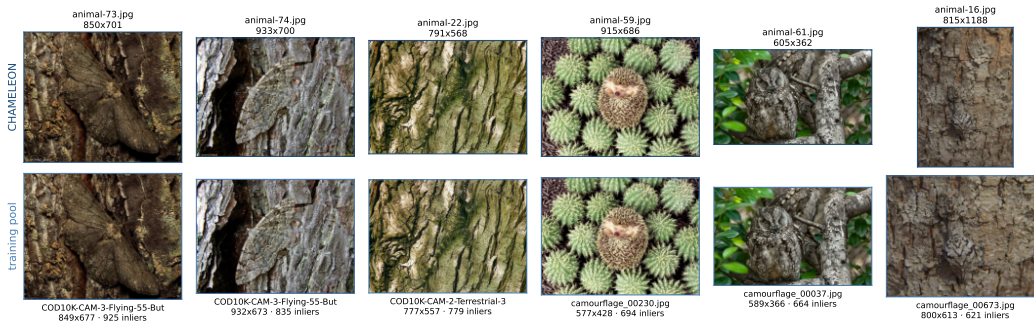


Figure 4: Six of the 9 CHAMELEON images that the released same-dimension detector reports UNCHECKED and the geometric extension matches: the author-sourced CHAMELEON image (top) and its partner in the training pool (bottom), with each pair’s RANSAC inlier count. *Selection rule, applied not curated*: the six highest inlier counts among the newly matched. Note the differing dimensions in every pair — that is why a same-dimension check cannot see them, and why the 41 of Section 5.2 was a lower bound. A tenth candidate cleared the inlier threshold but failed the residual test and is excluded; its homography is degenerate (Appendix H). *Source*: rebuild/FINAL\_AUDIT/out/final\_verdicts.json.

*One candidate cause we removed rather than kept.* A convenient explanation was available: that foreground information reaches the regenerated background only through a narrow low-dimensional summary. Examining the released generator’s architecture refutes that account as stated — four routes carry foreground information into the regenerated region, the narrow summary is one, and the dominant route bypasses it entirely. *This is resolved by source reading, not by measurement*: no experiment was run and no metric logged, and it withdraws one of our own candidate causes rather than establishing a result. Showing the channel is not narrow does not show its capacity is used, so we decline both the claim that the generator is not the binding constraint and that it is steerable.

## F BENCHMARK CONTAMINATION AUDIT CORRECTIONS

**Corrections to our own work.** An author-sourced re-audit reproduced 8 of 8 of our earlier measurements; its three mismatches correct our prior work, not the data. One declared threshold is left *failing* rather than reworded: we required every confirmed pair to differ in its JPEG quantization table and compared with an inequality that scored a *missing* table as a differing one — absence of evidence read as evidence — because two files are PNGs carrying a .jpg extension. Corrected, 41 of 41 pairs still carry independent re-encoding evidence, 40 by quantization table and one by container format. Separately, CHAMELEON was never a sanctioned endpoint in the repository we built on; it was our own plan that promoted it, so the framing here is not that we found a defect in someone else’s evaluation.

## G ENDPOINT-AVAILABLE SIGNAL VALIDATION

**A correction to our own measurement.** Our first measurement correlated this signal *on the labelled endpoint* against endpoint error — not the quantity the pipeline allocates by. Measured on the signal actually available, correlation with pixel error falls by +0.2100 and +0.2470, from +0.8695 and +0.8754 to +0.6595 and +0.6284. A signal validated where labels exist overstates what is usable where they do not.

## H CHAMELEON CONTAMINATION: CALIBRATION, CONTROLS AND ADJUDICATION

**Tier A — the released same-dimension detector.** Images are grouped by exact pixel dimensions, since a re-encode preserves them. Within a group each image is reduced to a  $32 \times 32$  greyscale

descriptor, pairs are shortlisted by descriptor RMS distance at 14.0, and survivors are decoded at full resolution and confirmed when the mean absolute difference over RGB falls below 6/255. We deliberately do *not* contrast-normalise the descriptor: normalising destroys the scale that separates a re-encode from a merely similar photograph, and it is what made our own first pass report 10 of 76 rather than 41. Sweeping the tolerance over 1, 2, 3, 5, 6 gives 11, 26, 37, 40 and 41 — the count saturates. The detector passes an eight-assertion synthetic known-answer test that uses no repository data.

**Tier B — calibration of the geometric extension.** SIFT correspondences, Lowe ratio 0.75, filtered by a RANSAC homography at 4 px. Thresholds are fixed on data whose answer is already committed, before the unchecked images are scored: positives are the 41 Tier-A pairs, negatives 4000 random CHAMELEON×training pairs.

|                                      |                        |
|--------------------------------------|------------------------|
| Inliers on the 41 confirmed pairs    | 525–1174 (median 1129) |
| Inliers on 4000 random pairs         | max 8                  |
| Inliers anywhere in the NC4K control | max 19                 |
| <b>Operating point</b>               | 20                     |
| Recall on the knowns at that point   | 41/41                  |
| NC4K flagged at that point           | 0 of 120               |

The operating point is one inlier above the largest count *any* negative set produced, which is the NC4K maximum of 19 rather than the random-pair maximum of 8. We also ran resolution-normalised global matching as a second instrument; no unchecked image passed it (lowest normalised RMS 16.7 against a calibrated threshold of 1.02), so every Tier-B match rests on the key-point stage alone.

**Adjudication — from “the keypoints agree” to “the same photograph”.** An inlier count shows that local patches agree on a geometric model. Two exposures of one static scene agree on a homography as well, so the inlier count alone cannot carry the claim that these files are the same photograph. For each candidate we recover the homography, warp the partner into the CHAMELEON frame and measure the residual over the valid overlap, and we decompose the homography into scale, rotation, shear and perspective. Run on 8 of the 41 known pairs the instrument returns their recorded mean absolute difference to three decimals ( $0.603 \rightarrow 0.603$ ,  $5.512 \rightarrow 5.514$ ) at  $H = I$ , which is what licenses reading its output on the rest.

| CHAMELEON | Inliers | Residual                                                                                     | Corr. | Scale       | Partner pool |
|-----------|---------|----------------------------------------------------------------------------------------------|-------|-------------|--------------|
| animal-16 | 2015    | 3.76                                                                                         | 0.99  | 1.02        | CAMO         |
| animal-73 | 3332    | 3.93                                                                                         | 1.00  | 1.00 (crop) | COD10K-train |
| animal-74 | 3403    | 4.17                                                                                         | 1.00  | 1.00 (crop) | COD10K-train |
| animal-22 | 3097    | 5.84                                                                                         | 1.00  | 1.00 (crop) | COD10K-train |
| animal-72 | 396     | 6.60                                                                                         | 0.99  | 6.36        | CAMO         |
| animal-61 | 1456    | 8.92                                                                                         | 0.97  | 0.98        | CAMO         |
| animal-43 | 263     | 9.54                                                                                         | 0.96  | 0.53        | CAMO         |
| animal-66 | 860     | 10.72                                                                                        | 0.97  | 0.51/0.39   | COD10K-train |
| animal-59 | 1017    | 12.11                                                                                        | 0.97  | 1.52        | CAMO         |
| animal-7  | 25      | <b>rejected</b> — shear $-86.5^\circ$ , scale 0.24 vs 1.24; no homography at full resolution |       |             |              |

Residual is mean  $|\cdot|$  over the overlap on a 0–255 scale. Shear and perspective are indistinguishable from zero for all nine. animal-7 was the only candidate below the calibration band of 525–1174 inliers, clearing the operating point by 5 while the NC4K negatives reached 19; its homography collapses the frame towards a line, and it is excluded.

**Scope of the exact-hash zero.** The 0 we report is measured against the training pool. Across benchmarks it is not zero: animal-17, animal-25 and animal-60 are byte-identical to CAMO-250 images. Those files are absent from the training pool and so do not enter the contamination count, but CAMO-250 is the split this repository selects checkpoints on.

**What remains uncouned.** Sixteen images have no partner at either tier and are reported UNCHECKED, not clean. Different photographs of one specimen, montages and heavy colour edits are not detected and are not claimed. Every figure here is a lower bound.

## I CHAMELEON MASK RELEASES AND DETECTOR VALIDATION

The two circulating CHAMELEON ground-truth releases are different annotations, unrelated to contamination or to any method. They disagree in polarity, giving a mean IoU of 0.0 as stored and 0.6932 after polarity alignment; only 27 of 76 masks are exactly identical after alignment and 40 of 76 reach  $\text{IoU} \geq 0.9$ . On identical predictions the choice of release alone moves MAE by  $2.7\times$  (0.2196 against the author-sourced masks, 0.0816 against the repackaged ground truth). Any CHAMELEON column is therefore not comparable across papers unless the release is named.

**Detector self-test.** The released detector passes an eight-assertion synthetic known-answer test using no repository data, reproduces the canonical 41 pairs and 41 names exactly, and surfaced one additional pair — the second partner of the one image that has two.

## J SECONDARY-ENDPOINT SENSITIVITY CHECK

**A threshold that passed, and was refused.** On the secondary endpoint,  $\Delta(A0 \rightarrow C10)$  for SINet is  $+0.0109$  with sign consistent 3/3, against a bar of 0.0083: the frozen rule returns REAL EFFECT. We do not report it as one, on two independent grounds each sufficient by itself. NC4K is the secondary endpoint and decides nothing, and the comparison is against A0, which is both confounded in pool size and the least stable arm in the campaign. We record it prominently *because* it is the tempting one.

## K ARM ANCHORING AND BASELINE STABILITY

**Where these arms sit against the published numbers.** The published configuration corresponding to arm C10 reports  $S_\alpha = 0.7136$  on this endpoint; our C10 measures 0.7185. A0 is the *unpadded* baseline and is not that configuration: its reproduction gate was on seed 42, which measured 0.7151 against this repository’s committed 0.7172, inside the declared  $3\sigma = 0.0107$ . The A0 arm *mean* of 0.7010 sits lower because A0 is the least stable arm, spanning 0.0334 across seeds on SINet and 0.0211 on SINet-v2. We offer a mechanism rather than leaving it unexplained: A0 holds 4447 images where every other arm holds 5447 under an identical pinned step count, so it samples a different mixture and should carry higher seed variance — and the effect reproduces on both architectures, which argues for a property rather than a broken run. We did not test this, and state it as a hypothesis.

## L SUMMARY OF CAUSES AND SCOPE

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Table 6: Per-run metrics at full recorded precision for every training run in this work — the two frozen campaigns and the mean-teacher positive control — on both endpoints. Arms B and C10 appear once; they were re-scored, not re-trained, for the falsification campaign and reproduced their committed values exactly at six decimal places. *Source:* `rebuild/ABC/out/abc_metrics.csv` and `out/t2/abc_metrics.csv`; blocks EXP ABC #3 and EXP T2 #3.

| arch    | arm   | seed | endpoint | $S_\alpha$ | MAE      |
|---------|-------|------|----------|------------|----------|
| SINet   | A0    | 42   | COD10K   | 0.715086   | 0.074960 |
| SINet   | A0    | 43   | COD10K   | 0.681706   | 0.077044 |
| SINet   | A0    | 45   | COD10K   | 0.706058   | 0.077347 |
| SINet   | A2    | 42   | COD10K   | 0.705507   | 0.075994 |
| SINet   | A2    | 43   | COD10K   | 0.708334   | 0.078489 |
| SINet   | A2    | 45   | COD10K   | 0.709196   | 0.080858 |
| SINet   | B     | 42   | COD10K   | 0.711554   | 0.076085 |
| SINet   | B     | 43   | COD10K   | 0.713632   | 0.076278 |
| SINet   | B     | 45   | COD10K   | 0.715155   | 0.075917 |
| SINet   | C10   | 42   | COD10K   | 0.717165   | 0.073184 |
| SINet   | C10   | 43   | COD10K   | 0.723034   | 0.075136 |
| SINet   | C10   | 45   | COD10K   | 0.715243   | 0.078355 |
| SINet   | CINV  | 42   | COD10K   | 0.722612   | 0.073910 |
| SINet   | CINV  | 43   | COD10K   | 0.717343   | 0.077323 |
| SINet   | CINV  | 45   | COD10K   | 0.718084   | 0.073607 |
| SINet   | CSHUF | 42   | COD10K   | 0.718602   | 0.074840 |
| SINet   | CSHUF | 43   | COD10K   | 0.716974   | 0.077678 |
| SINet   | CSHUF | 45   | COD10K   | 0.715291   | 0.079340 |
| SINet   | MT    | 42   | COD10K   | 0.699633   | 0.093377 |
| SINet   | MT    | 43   | COD10K   | 0.699167   | 0.087571 |
| SINet   | MT    | 45   | COD10K   | 0.697503   | 0.091322 |
| SINetv2 | A0    | 42   | COD10K   | 0.688813   | 0.091691 |
| SINetv2 | A0    | 43   | COD10K   | 0.678740   | 0.095147 |
| SINetv2 | A0    | 45   | COD10K   | 0.699883   | 0.085756 |
| SINetv2 | A2    | 42   | COD10K   | 0.691914   | 0.089186 |
| SINetv2 | A2    | 43   | COD10K   | 0.688977   | 0.094524 |
| SINetv2 | A2    | 45   | COD10K   | 0.695414   | 0.087455 |
| SINetv2 | B     | 42   | COD10K   | 0.698340   | 0.092242 |
| SINetv2 | B     | 43   | COD10K   | 0.689481   | 0.089883 |
| SINetv2 | B     | 45   | COD10K   | 0.697385   | 0.087132 |
| SINetv2 | C10   | 42   | COD10K   | 0.692321   | 0.093395 |
| SINetv2 | C10   | 43   | COD10K   | 0.696328   | 0.090749 |
| SINetv2 | C10   | 45   | COD10K   | 0.695360   | 0.089182 |
| SINetv2 | CINV  | 42   | COD10K   | 0.697008   | 0.088438 |
| SINetv2 | CINV  | 43   | COD10K   | 0.696331   | 0.087293 |
| SINetv2 | CINV  | 45   | COD10K   | 0.697605   | 0.090189 |
| SINetv2 | CSHUF | 42   | COD10K   | 0.701269   | 0.087506 |
| SINetv2 | CSHUF | 43   | COD10K   | 0.694490   | 0.087305 |
| SINetv2 | CSHUF | 45   | COD10K   | 0.696218   | 0.089485 |
| SINet   | A0    | 42   | NC4K     | 0.765121   | 0.085752 |
| SINet   | A0    | 43   | NC4K     | 0.751545   | 0.088874 |
| SINet   | A0    | 45   | NC4K     | 0.757700   | 0.090228 |
| SINet   | A2    | 42   | NC4K     | 0.760908   | 0.088385 |
| SINet   | A2    | 43   | NC4K     | 0.762933   | 0.087466 |
| SINet   | A2    | 45   | NC4K     | 0.765789   | 0.090892 |
| SINet   | B     | 42   | NC4K     | 0.764108   | 0.087897 |
| SINet   | B     | 43   | NC4K     | 0.768972   | 0.085001 |
| SINet   | B     | 45   | NC4K     | 0.767632   | 0.088175 |
| SINet   | C10   | 42   | NC4K     | 0.765296   | 0.086273 |
| SINet   | C10   | 43   | NC4K     | 0.771241   | 0.085305 |
| SINet   | C10   | 45   | NC4K     | 0.770503   | 0.086622 |
| SINet   | CINV  | 42   | NC4K     | 0.774697   | 0.083144 |
| SINet   | CINV  | 43   | NC4K     | 0.774114   | 0.084030 |
| SINet   | CINV  | 45   | NC4K     | 0.768136   | 0.084570 |
| SINet   | CSHUF | 42   | NC4K     | 0.766713   | 0.085734 |
| SINet   | CSHUF | 43   | NC4K     | 0.767983   | 0.085825 |
| SINet   | CSHUF | 45   | NC4K     | 0.767229   | 0.089734 |
| SINet   | MT    | 42   | NC4K     | 0.752869   | 0.108067 |
| SINet   | MT    | 43   | NC4K     | 0.746708   | 0.101192 |
| SINet   | MT    | 45   | NC4K     | 0.749400   | 0.104323 |
| SINetv2 | A0    | 42   | NC4K     | 0.745691   | 0.092348 |
| SINetv2 | A0    | 43   | NC4K     | 0.736067   | 0.099838 |

Table 7: Which evaluation columns can be reported as independent measurements. Every rate is still a *lower bound*: different photographs of one specimen, montages and heavy colour edits are not detected, and an image with no match is unchecked, not clean. The CHAMELEON row combines the released same-dimension detector’s 41 with 9 rescaled or cropped copies recovered by the geometric extension and confirmed by post-warp residual; the other three rows are same-dimension only, so their unchecked shares are larger than stated here. In the exact-hash column 0 is a measured zero against the *training pool*, “—” not applicable; it is not a claim about overlap between benchmarks (Section 5.2). <sup>‡</sup>the nearest-neighbour gap is evidence for the same-dimension tier of 41, not for the combined figure. *Source*: CLEAN\_PROTOCOL.md; EXP D2 #5, D2R #4, D2\_NC4K #2; extension and adjudication in rebuild/FINAL\_AUDIT/out/final\_verdicts.json.

| Endpoint         | Contaminated         | Exact hash | Unchecked | Nearest-neighbour gap                  | Verdict                 |
|------------------|----------------------|------------|-----------|----------------------------------------|-------------------------|
| COD10K-test      | 2/2026 (0.1%)        | —          | 524/2026  | none                                   | reportable <sup>†</sup> |
| NC4K             | 1/4121 (0.0%)        | —          | 2406/4121 | none                                   | reportable <sup>†</sup> |
| <b>CHAMELEON</b> | <b>50/76 (65.8%)</b> | <b>0</b>   | 16/76     | 41 below 5.51, next 40.58 <sup>‡</sup> | <b>not reportable</b>   |
| CAMO             | 4/250 (1.6%)         | —          | 155/250   | none                                   | selection set only      |

<sup>†</sup> Measured against on-disk copies, not author-sourced ones — the same weakness the CHAMELEON re-audit closes. That re-audit is *owed* for these rows; until then their rates are provisional.

Table 8: Post-hoc interval estimates for the gaps this paper turns on, on COD10K-test  $S_\alpha$ . *These decide nothing*: the verdicts remain the pre-registered  $2\hat{\sigma}$  rule of Section 3.4, which reports no  $p$ -value at  $n = 3$ . Intervals are Welch (unequal-variance) on the two arms actually compared, so they are not inflated by A0. TOST is run at the SESOI  $\delta = 0.005$ ; the final column gives the smallest  $\delta$  at which each comparison *could* have concluded equivalence, which is the honest positive statement the data supports. *Source*: rebuild/DIAG/out/diag\_tost.json.

| Arch.    | Gap         | $\Delta$ | 90% CI             | 95% CI             | smallest $\delta$ |
|----------|-------------|----------|--------------------|--------------------|-------------------|
| SINet    | C10 – B     | +0.0050  | [−0.0012, +0.0113] | [−0.0035, +0.0136] | 0.0113            |
| SINet-v2 | C10 – B     | −0.0004  | [−0.0079, +0.0071] | [−0.0107, +0.0099] | 0.0079            |
| SINet    | C10 – CSHUF | +0.0015  | [−0.0048, +0.0078] | [−0.0072, +0.0102] | 0.0078            |
| SINet    | C10 – CINV  | −0.0009  | [−0.0072, +0.0054] | [−0.0092, +0.0075] | 0.0072            |
| SINet-v2 | C10 – CSHUF | −0.0027  | [−0.0080, +0.0027] | [−0.0099, +0.0045] | 0.0080            |
| SINet-v2 | C10 – CINV  | −0.0023  | [−0.0056, +0.0010] | [−0.0070, +0.0024] | 0.0056            |

No comparison reaches equivalence at  $\delta = 0.005$ : at  $n = 3$  per arm a TOST has almost no power, and reporting that plainly is more useful than reporting the  $p$ -values. What the intervals *do* establish is stated in Section 4.

Table 9: The ten most-funded clusters, and what the budget is actually aimed at. The target pool is 24.8% CAMO, but 51.5% of the generation budget lands on CAMO images — a  $2.08\times$  enrichment ( $2.14\times$  under the committed assignment). The two largest quotas take 33.5% of the budget, are 90.2% CAMO, and between them hold 7 of the 2026 primary-endpoint images. *Source:* rebuild/DIAG/out/diag\_clusters.json.

| Cluster                                                            | Budget | % of $B$ | $n_{\text{target}}$ | CAMO % | $n_{\text{test}}$ | $es_j$       |
|--------------------------------------------------------------------|--------|----------|---------------------|--------|-------------------|--------------|
| 30                                                                 | 194    | 19.4     | 21                  | 95.2   | 2                 | 0.0707       |
| 34                                                                 | 141    | 14.1     | 81                  | 85.2   | 5                 | 0.0674       |
| 63                                                                 | 74     | 7.4      | 18                  | 66.7   | 3                 | 0.0609       |
| 13                                                                 | 67     | 6.7      | 57                  | 17.5   | 24                | 0.0599       |
| 23                                                                 | 29     | 2.9      | 43                  | 41.9   | 20                | 0.0515       |
| 73                                                                 | 26     | 2.6      | 39                  | 46.2   | 16                | 0.0505       |
| 65                                                                 | 24     | 2.4      | 64                  | 51.6   | 23                | 0.0496       |
| 19                                                                 | 23     | 2.3      | 93                  | 8.6    | 66                | 0.0490       |
| 0                                                                  | 22     | 2.2      | 57                  | 1.8    | 31                | 0.0485       |
| 64                                                                 | 19     | 1.9      | 141                 | 29.1   | 74                | 0.0471       |
| <i>Budget-weighted totals against the pool they are drawn from</i> |        |          |                     |        |                   |              |
| CAMO share of the target pool                                      |        |          |                     |        |                   | 0.248        |
| CAMO share of the <i>budget</i>                                    |        |          |                     |        |                   | <b>0.515</b> |
| Endpoint images per unit budget, against uniform                   |        |          |                     |        |                   | 0.722        |

Table 10: Arm differences on *boundary* metrics, COD10K-test, same three seeds, same pooled- $\hat{\sigma}$  rule. **Post-hoc: these metrics were not pre-registered and no campaign verdict is re-decided.** On SINet the concentration gap C10 – B clears its bar — and so does CINV – B, the *anti*-targeted arm, by as much; the two direction contrasts stay within noise. SINet-v2 resolves none of it, and that disagreement is reported as a disagreement. *Source:* rebuild/DIAG/out/diag\_subgroups.json.

| Arch.    | Metric       | $2\hat{\sigma}$ | C10 – B            | CINV – B           | C10 – CSHUF | C10 – CINV  |
|----------|--------------|-----------------|--------------------|--------------------|-------------|-------------|
| SINet    | Boundary IoU | 0.0101          | +0.0138 <b>3/3</b> | +0.0135 <b>3/3</b> | +0.0038 2/3 | +0.0004 2/3 |
| SINet    | Boundary $F$ | 0.0151          | +0.0190 <b>3/3</b> | +0.0173 <b>3/3</b> | +0.0045 2/3 | +0.0017 2/3 |
| SINet    | IoU          | 0.0109          | +0.0112 <b>3/3</b> | +0.0127 <b>3/3</b> | +0.0019 2/3 | –0.0016 2/3 |
| SINet-v2 | Boundary IoU | 0.0062          | +0.0010 2/3        | +0.0025 2/3        | –0.0002 2/3 | –0.0015 3/3 |
| SINet-v2 | Boundary $F$ | 0.0065          | +0.0026 2/3        | +0.0050 3/3        | –0.0002 2/3 | –0.0025 3/3 |
| SINet-v2 | IoU          | 0.0100          | –0.0003 2/3        | +0.0006 2/3        | –0.0013 2/3 | –0.0010 2/3 |

Bold marks a gap exceeding  $2\hat{\sigma}$  with 3/3 sign consistency. Counts after each  $\Delta$  are sign-consistency across the three seeds. **Threshold robustness (SINet, Boundary IoU):** at binarisation thresholds 0.4/0.5/0.6,  $\Delta(\text{C10} - \text{B})$  is  $1.39/1.37/1.23\times$  bar and  $\Delta(\text{CINV} - \text{B})$  is  $1.25/1.34/1.26\times$ , both 3/3 at every threshold, while  $\Delta(\text{C10} - \text{CSHUF})$  and  $\Delta(\text{C10} - \text{CINV})$  never exceed  $0.52\times$  and  $0.14\times$ . The pattern is not an artifact of the threshold.

Table 11: Every cause, its measured quantity, and how far it carries. Causes marked *generalizes* concern the target data or the uncertainty signal and travel beyond this instantiation; those marked *specific to this setup* are properties of this loop or this generator. The scope column, not the magnitude column, determines how far a row may be carried. *Sources, by row:* FINAL\_RESULTS.md Tables 8.2, 8.1, 9.2, 9.4, 11.1, 12.1, 5.1, 7.1 and 10.4; blocks EXP B1 #3 and #4, EXP C1 #4, EXP T2 #3, EXP T2C, EXP ABC #2, EXP D1, EXP A3, in results/REBUILD\_LOG.txt. The conditioning row has no log block by construction.

| Cause                                                 | Measured quantity                                            | Scope                                      | Carries to                                                        |
|-------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------|-------------------------------------------------------------------|
| No cluster structure to allocate over                 | best silhouette 0.1600; 0.1465, 0.0568 elsewhere             | generalizes                                | any method allocating over clusters of such a distribution        |
| Membership is not a robust label                      | 5.4% of memberships move under a preprocessing change        | generalizes                                | a floor beneath which an effect cannot be attributed to targeting |
| Signal adds nothing beyond concentration, in geometry | +0.0073 of a $d$ against its own shuffle, 13/20              | generalizes                                | any uncertainty-guided acquisition: the shuffle is the control    |
| ... and nothing resolvable on trained accuracy        | 12/12 WITHIN NOISE, 3.24 $\times$ tighter bar                | generalizes                                | this signal class carries no resolvable directional information   |
| Targeting buys proximity, not coverage                | effective-rank ratio 0.53–0.64, 20/20                        | generalizes                                | concentration narrows the set rather than extending its reach     |
| Pixel error predicted better than structural error    | ordering passes 8/8 rows, 3 signals $\times$ 2 architectures | generalizes                                | uncertainty-guided allocation for this task in general            |
| Endpoint-measured signal overstates itself            | $\rho$ drops +0.2100, +0.2470                                | generalizes                                | any signal validated where labels exist                           |
| Optimisation budget is pinned                         | 253 / 127 steps, both rounds, every run                      | specific to this setup                     | nothing — a property of this loop                                 |
| Foreground pool is exhausted                          | bijection 4447/4447; 0 outside, 0 unrendered                 | specific, but the scoping sentence travels | the scoping sentence any comparable null must carry               |
| Objects are copied, not generated                     | interior 5.603 / 9.633 vs background 70.802 / 70.555         | specific to this setup                     | the measurement design, reusable on any inpainting generator      |
| Generated distribution is narrow                      | recall 0.13–0.54 vs 0.71–0.90 for every real set             | specific to this setup                     | the design: a paired within-content coverage delta                |
| Conditioning channel is <i>not</i> the bottleneck     | four routes; source reading, no experiment                   | specific to this setup                     | nothing — a withdrawn cause                                       |